Highlights from two astronomy problems for Lagrange's equations: the Lagrange points and the cosmological redshift

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These notes use only what is in Taylor, *Classical Mechanics* [1], Sections 7.1, 7.3, 7.5, 7.6 and 7.8: Hamilton's principle, Lagrange's equations, constraints and degrees of freedom, a coordinate change that depends on time, ignorable coordinates, and "no t in $\mathcal{L}$ gives a conserved Hamiltonian." The first half is the restricted three-body problem as every celestial-mechanics book sets it up (e.g. Murray and Dermott [2], Chapter 3; the Lagrangian route is the one in Goldstein, Poole and Safko [3], Section 3.12); the second half, the cosmology, follows the standard textbook route for a particle in an expanding universe (Peebles [4], Chapter II). The references are collected at the end. Each problem is done straight through, in the order of Section 7.5's recipe, with Nailing down details (the numbers, a board problem, stability) collected at the end of its section; the redshift is also written up as a homework problem, statement and solutions (Section 3). Both halves are Newtonian throughout, and the places where general relativity would add something are flagged.

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1 The Lagrange points

This section does the problem the way every Chapter 7 problem is done, in the order of Section 7.5's recipe (p. 258) and Examples 7.4–7.7: set the problem up, write the Lagrangian with nothing left out, state the constraints, write the Lagrangian with the constraints in, take Lagrange's equations, read off what they say. Every step is here; the numbers, a board problem and the question of stability are collected at the end of the section, under "Nailing down details (p. 11)."

The problem. JWST sits about 1.5×10^6 km from Earth, on the far side from the Sun, at the point called L_2; Jupiter's Trojan asteroids cluster 60° ahead of and behind the planet, at L_4 and L_5. Lagrange found these five points in 1772 as special solutions of the three-body problem (the first Trojan, 588 Achilles, was not seen until 1906). In what sense is a point "at rest" when the Sun, the Earth and the point are all orbiting? In the frame that rotates with the two big bodies. Taylor does rotating frames with vectors in Chapter 9; here the same physics drops out of a Lagrangian in three lines, by the time-dependent coordinate change of Section 7.5.

The idea in one paragraph. The inertial coordinates (X, Y) are the ordinary, time-independent ones; the trouble is that in them the Lagrangian carries t explicitly, because the primaries move and so $U(X, Y, t)$ changes in time, and neither of Chapter 7's tests (an ignorable coordinate, Section 7.6; no explicit t, Section 7.8) finds a conserved quantity. The move is to change to rotating coordinates (x, y), which are themselves time-dependent functions of (X, Y), the kind of generalized coordinate Section 7.3 allows and Example 7.6's spinning hoop uses. In them the primaries sit still, so $U(x, y)$ has no t, and the Hamiltonian is conserved: that is the Jacobi integral.[1] The price of time-dependent coordinates is that they are not "natural" (Section 7.3), so T picks up terms linear in the velocities and a velocity-free term, and the conserved $\mathcal{H}$ is not $T + U$ (Section 7.8). The redshift problem of Section 2 is the mirror image: the comoving coordinate $x = r/a(t)$ is also a time-dependent change, and it makes x ignorable, but t survives in $\mathcal{L}$ through $a(t)$, so there the Hamiltonian is not conserved. One problem gains a conserved $\mathcal{H}$ from the change of coordinates; the other does not.

The result, first.

$$\boxed{\mathcal{L} = \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + m\Omega\,(x\dot{y} - y\dot{x}) + \tfrac{1}{2}m\Omega^2(x^2 + y^2) - U(x, y)} \tag{1.1}$$

with U the ordinary gravitational potential energy of m in the field of the two fixed primaries. Three things follow:

- Lagrange's equations are Newton's second law plus two extra forces, the centrifugal force $m\Omega^2(x, y)$ and the Coriolis force $2m\Omega(\dot{y}, -\dot{x})$: Chapter 9's results, for free.
- A body can sit still in the rotating frame exactly where the *effective potential* $U_{\text{eff}} \equiv U - \frac{1}{2}m\Omega^2(x^2 + y^2)$ has zero gradient. There are five such points ("Polar coordinates" in Nailing down details shows why exactly five). Two of them, L_4 and L_5, form equilateral triangles with M_1 and M_2, for any mass ratio; the other three lie on the line through the primaries.
- $\mathcal{L}$ has no explicit t, so $\mathcal{H}$ is conserved. It is $\frac{1}{2}mv_{\text{rot}}^2 + U_{\text{eff}}$ (with $v_{\text{rot}}^2 = \dot{x}^2 + \dot{y}^2$, the speed squared in the rotating frame), called the *Jacobi integral*, and it is *not* $T + U$.

Step 1: set up the problem. Two bodies M_1 and M_2 (typically called "the primaries"), a distance R apart, go around their centre of mass (CM) in circles at angular speed Ω; a third body of mass

[1] Named for C. G. J. Jacobi; background and the usual conventions are in [5].

m moves in their gravity. Put the origin at the CM and use inertial Cartesian coordinates (X, Y, Z) with the primaries' orbit in the plane $Z = 0$. "Origin at the CM" means $M_1\,\mathbf{r}_1 + M_2\,\mathbf{r}_2 = 0$, i.e. $M_1 r_1 = M_2 r_2$ with $r_1 = |\,\mathbf{r}_1|$ and $r_2 = |\,\mathbf{r}_2|$, the two bodies on opposite sides of the CM. The sum of the two distances is $r_1 + r_2 = R$. Put $r_2 = R - r_1$ into $M_1 r_1 = M_2 r_2$:

$$\begin{aligned} M_1 r_1 &= M_2(R - r_1) \\ (M_1 + M_2)\, r_1 &= M_2 R && \text{(the } M_2 r_1 \text{ moved to the left)} \\ r_1 &= \frac{M_2 R}{M_1 + M_2}, && \text{and likewise} \quad r_2 = \frac{M_1 R}{M_1 + M_2}. \end{aligned} \tag{1.2}$$

It will be handy to have a name for the mass ratio: with $\mu \equiv M_2/(M_1 + M_2)$, these are $r_1 = \mu R$ and $r_2 = (1 - \mu)R$, since $M_1/(M_1 + M_2) = 1 - \mu$. For the Sun and the Earth $\mu = 3.00 \times 10^{-6}$; for the Earth and the Moon, 0.0122. Each primary goes around the CM on a circle of its own radius, r_1 or r_2, at the common angular speed Ω, so its polar angle is Ωt plus a constant; choose $t = 0$ when M_2 is on the positive X axis, so M_2's angle is Ωt, and M_1, always on the opposite side of the CM, is at $\Omega t + \pi$, which flips the sign of both components. In Cartesian components the primaries are at

$$\mathbf{r}_1(t) = r_1(-\cos\Omega t,\, -\sin\Omega t,\, 0), \qquad \mathbf{r}_2(t) = r_2(\cos\Omega t,\, \sin\Omega t,\, 0). \tag{1.3}$$

Newton's second law for M_1 fixes the rate: its acceleration is $\Omega^2 r_1$ toward the CM (uniform circular motion) and the only force on it is M_2's pull GM_1M_2/R^2, also toward the CM, so

$$\begin{aligned} M_1\Omega^2 r_1 &= \frac{GM_1M_2}{R^2} && \text{(mass} \times \text{acceleration} = \text{force)} \\ M_1\Omega^2\,\frac{M_2 R}{M_1 + M_2} &= \frac{GM_1M_2}{R^2} && (r_1 \text{ from (1.2)}) \\ \Omega^2 &= \frac{G(M_1 + M_2)}{R^3}. && \text{(cancel } M_1M_2\text{, multiply by } (M_1 + M_2)/R) \end{aligned} \tag{1.4}$$

This is Kepler's third law for the pair.

Step 2: the Lagrangian, without worrying about constraints yet. The body m has three Cartesian coordinates and (so far) no constraint. So, $\mathcal{L} = T - U$ with

$$\mathcal{L} = \tfrac{1}{2}m\big(\dot X^2 + \dot Y^2 + \dot Z^2\big) - U(X, Y, Z, t), \qquad U = -\frac{GmM_1}{|\,\mathbf{r} - \mathbf{r}_1(t)|} - \frac{GmM_2}{|\,\mathbf{r} - \mathbf{r}_2(t)|}, \tag{1.5}$$

where $|\,\mathbf{r} - \mathbf{r}_i(t)| = \sqrt{(X - X_i(t))^2 + (Y - Y_i(t))^2 + Z^2}$ and (X_i, Y_i) are the components in (1.3). Lagrange's equations for X, Y and Z are Newton's second law with the two attractions: for X, $\partial\mathcal{L}/\partial\dot X = m\dot X$ and $\partial\mathcal{L}/\partial X = -\partial U/\partial X$, so $\frac{d}{dt}\partial\mathcal{L}/\partial\dot X = \partial\mathcal{L}/\partial X$ reads $m\ddot X = -\partial U/\partial X$, and likewise for Y and Z; together, $m\,\ddot{\mathbf{r}} = -\boldsymbol{\nabla}U$.

To see that $-\boldsymbol{\nabla}U$ is the two attractions, write $d_i = |\,\mathbf{r} - \mathbf{r}_i(t)|$ for the two distances, so that $U = -GmM_1/d_1 - GmM_2/d_2$. Then

$$\begin{aligned} -\boldsymbol{\nabla}U &= -\Big(\frac{\partial U}{\partial X}, \frac{\partial U}{\partial Y}, \frac{\partial U}{\partial Z}\Big) \\ &= GmM_1\Big(\frac{\partial}{\partial X}\frac{1}{d_1}, \frac{\partial}{\partial Y}\frac{1}{d_1}, \frac{\partial}{\partial Z}\frac{1}{d_1}\Big) + GmM_2\Big(\frac{\partial}{\partial X}\frac{1}{d_2}, \frac{\partial}{\partial Y}\frac{1}{d_2}, \frac{\partial}{\partial Z}\frac{1}{d_2}\Big) \\ &= GmM_1\,\boldsymbol{\nabla}\frac{1}{d_1} + GmM_2\,\boldsymbol{\nabla}\frac{1}{d_2}, \end{aligned}$$

so what we need are the derivatives of $1/d_1$ and $1/d_2$. Take the X derivative of $1/d_1$. Differentiating the square root $d_1 = \sqrt{(X-X_1)^2 + (Y-Y_1)^2 + Z^2}$ gives $\partial d_1/\partial X = (X - X_1)/d_1$, so by the chain rule

$$\frac{\partial}{\partial X}\frac{1}{d_1} = -\frac{1}{d_1^2}\frac{\partial d_1}{\partial X} = -\frac{X - X_1}{d_1^3};$$

the Y and Z derivatives come the same way, and we get

$$\boldsymbol{\nabla}\frac{1}{d_1} = -\frac{(X - X_1,\ Y - Y_1,\ Z)}{d_1^3} = -\frac{\mathbf{r} - \mathbf{r}_1}{d_1^3} = \frac{\mathbf{r}_1 - \mathbf{r}}{d_1^3}.$$

Likewise, $\boldsymbol{\nabla}(1/d_2) = (\mathbf{r}_2 - \mathbf{r})/d_2^3$. So

$$-\boldsymbol{\nabla}U = GmM_1\,\frac{\mathbf{r}_1 - \mathbf{r}}{d_1^3} + GmM_2\,\frac{\mathbf{r}_2 - \mathbf{r}}{d_2^3} = \frac{GmM_1}{d_1^2}\,\frac{\mathbf{r}_1 - \mathbf{r}}{d_1} + \frac{GmM_2}{d_2^2}\,\frac{\mathbf{r}_2 - \mathbf{r}}{d_2}$$

which we can read as two forces of magnitude GmM_i/d_i^2 along the unit vectors $(\mathbf{r}_i - \mathbf{r})/d_i$ toward the primaries. Neither of Chapter 7's tests finds a conserved quantity as it stands: t appears explicitly through $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$, so Section 7.8's $\mathcal{H}$ is not constant, and no coordinate is ignorable (Section 7.6), since U depends on all three.

Step 3: the constraints. We're holding two things fixed here. Let's look at what "held fixed" means in each case.

(i) *The motion of the primaries prescribed (fixed)* by construction. $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$ are given functions of t, not coordinates to be varied. That sounds like cheating *and it is!* The problem is that we have to cheat. The full problem, as stated to this point, is the three-body problem. Nine coordinates, nine coupled equations of motion. Newton, Euler, Lagrange, Poincaré and many others spent, between them, two centuries working on the three-body problem. Eventually Bruns (1887) [6] and Poincaré (1890) [7] showed that it has no conserved quantities beyond the ten everyone already knew (energy, momentum, angular momentum and the centre of mass), which is what a closed-form solution would need; Poincaré's memoir is also where chaos was first seen. That's cool, maybe we could get away with a series solution. Sundman (1912) [8] found one, and proved that it converges. Unfortunately, it converges terribly slowly: Belorizky estimated in 1930 [9] that reaching the accuracy of astronomical observations would take about $10^{8\,000\,000}$ terms! (Diacu [10] tells the whole story in five readable pages.) So, apart from very special cases, the three-body problem has no usable solution in closed form, and we cheat by holding something fixed.

Well, m is, in our case, tiny compared to M_1 and M_2, so it's barely going to affect their motion at all. And hence, we decide to hold the motion of the primaries fixed, while only varying the motion of m. This is called the *restricted* three-body problem.[2] We note that it's the same sort of restriction used in Example 7.6 (the bead on a spinning hoop) where the hoop's angle ωt is prescribed and only the bead's θ is a coordinate. It is a constraint on the system, not on m: it removes the six coordinates of the primaries from the problem and is what put t into U.

[2]The same approximation runs through quantum mechanics and chemistry. The hydrogen molecular ion H_2^+ with clamped nuclei is the quantum version: one light particle in the field of two heavy centres whose positions are prescribed, not solved for. There the justification is that the nuclei are slow (the Born–Oppenheimer separation of time scales); here it is that the third body's pull on the primaries is of order m/M and is dropped. Same approximation, two reasons to believe it. One more parallel: with the rotation switched off, the classical problem is Euler's problem of two fixed centres, which is exactly solvable, just as H_2^+ separates in prolate spheroidal coordinates. What makes the rotating problem hard is the Coriolis force, the velocity-dependent term in the equations of motion (1.11) (Step 5).

(ii) *We set* $Z = 0$. This is not a constraint in Section 7.3's sense, since no wire or surface enforces it. Rather, it is two choices. The first is a choice of coordinates: the motion of our primaries is confined to a plane, and we're just choosing that to be the XY plane. The second is a choice of initial conditions for m, which the equations of motion respect. We can spell this out a bit more explicitly if you're skeptical: U is even in Z (both distances contain Z only as Z^2), so

$$\frac{\partial U}{\partial Z} = \frac{GmM_1\,Z}{|\,\mathbf{r} - \mathbf{r}_1|^3} + \frac{GmM_2\,Z}{|\,\mathbf{r} - \mathbf{r}_2|^3}, \qquad \text{so on } Z=0\text{:} \qquad \frac{\partial U}{\partial Z} = 0 \quad \text{and} \quad m\ddot{Z} = -\frac{\partial U}{\partial Z} = 0.$$

So, a body that starts in the plane with $\dot{Z} = 0$ stays in it, and we look only at those motions. What about our little mass m wanting to leave the plane? Our goal is to find the places where it sits still with respect to the primaries, and those are all in the plane: $\partial U/\partial Z$ above is Z times a positive factor, so it vanishes only at $Z = 0$, and a body held off the plane is pulled back toward it. Nothing is lost by staying in the plane.

So once the primaries are prescribed, m has three degrees of freedom; restricting to the planar motions of (ii) removes Z and does nothing to X and Y. Two generalized coordinates remain.

Step 4: the Lagrangian with the constraints in, in good coordinates. With $Z = 0$ the Lagrangian (1.5) loses its $\dot{Z}$ term and U loses Z:

$$\mathcal{L} = \tfrac{1}{2}m\big(\dot{X}^2 + \dot{Y}^2\big) - U(X, Y, t) \tag{1.6}$$

and it's time to decide which two generalized coordinates to use. U only depends on t because the primaries move, so if we pick coordinates that move with the primaries, U will lose this t dependence. This may seem a bit unnatural, but Section 7.3 allows generalized coordinates whose relation to the position involves t (Example 7.6 uses one), and we opt for the rotating coordinates x and y given by

$$X = x\cos\Omega t - y\sin\Omega t, \qquad Y = x\sin\Omega t + y\cos\Omega t, \tag{1.7}$$

where (X, Y) are inertial: the (x, y) axes are the (X, Y) axes turned counterclockwise through the angle Ωt (the standard rotation matrix), so a point at fixed (x, y) goes around the origin at angular speed Ω, in step with the primaries. The other way round is the rotation through $-\Omega t$,

$$x = X\cos\Omega t + Y\sin\Omega t, \qquad y = -X\sin\Omega t + Y\cos\Omega t, \tag{1.8}$$

which is the form Section 7.3 has in mind, the generalized coordinates as functions of the position and t. We will not need it: we'll build $\mathcal{L}$ by putting (1.7) into (1.6) directly, which will give us an equation in terms of x, y and t.

In these new coordinates the primaries sit at $(x, y) = (-r_1, 0)$ and $(r_2, 0)$ for all time[3] You can see this intuitively (the whole reason we choose these co-moving coordinates was to fix the motion of the primaries), or you can see it explicitly by plugging $(x, y) = (r_2, 0)$ into the rotation: $X = r_2\cos\Omega t$, $Y = r_2\sin\Omega t$, which is (1.3), and $(-r_1, 0)$ likewise gives $\mathbf{r}_1(t)$. So the U of equation (1.5), the $U(X, Y, t)$ of (1.6), becomes

$$U(x, y) = -\frac{GmM_1}{\sqrt{(x + r_1)^2 + y^2}} - \frac{GmM_2}{\sqrt{(x - r_2)^2 + y^2}}, \tag{1.9}$$

[3] In terms of the mass ratio μ of Step 1, which Nailing down details (p. 11) uses, that is $(-\mu R, 0)$ and $((1-\mu)R, 0)$.

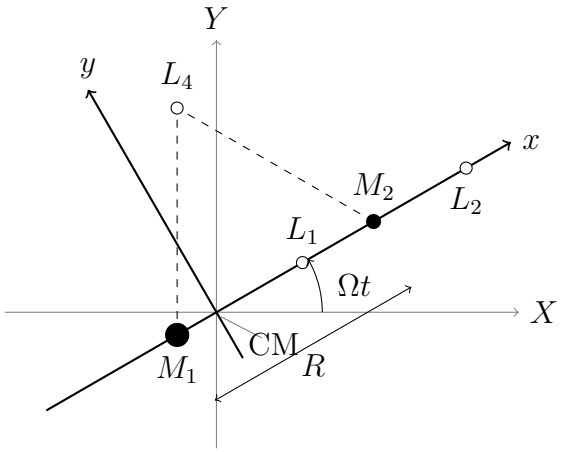


Figure 1: The inertial axes (X, Y) and the rotating axes (x, y), turned through Ωt about the Z axis, which runs through the CM and out of the page. The primaries sit on the x axis. L_1, L_2 and L_4 are three of the five Lagrange points, the places where m can sit still in the rotating frame; Step 6(a) and Nailing down details find them. L_4 is the point of the dashed equilateral triangle; L_1 and L_2 are drawn where they fall for this mass ratio ($\mu = 0.2$).

with no t. For the kinetic energy, we need $\dot{X}$ and $\dot{Y}$: differentiate (1.7) with the product rule (Ω is constant, so $\frac{d}{dt}\cos\Omega t = -\Omega\sin\Omega t$ and $\frac{d}{dt}\sin\Omega t = \Omega\cos\Omega t$):

$$\dot{X} = \dot{x}\cos\Omega t - \Omega x\sin\Omega t - \dot{y}\sin\Omega t - \Omega y\cos\Omega t,$$
$$\dot{Y} = \dot{x}\sin\Omega t + \Omega x\cos\Omega t + \dot{y}\cos\Omega t - \Omega y\sin\Omega t.$$

Collect the $\cos\Omega t$ and $\sin\Omega t$ terms:

$$\dot{X} = (\dot{x} - \Omega y)\cos\Omega t - (\dot{y} + \Omega x)\sin\Omega t,$$
$$\dot{Y} = (\dot{x} - \Omega y)\sin\Omega t + (\dot{y} + \Omega x)\cos\Omega t.$$

Square and add: the cross terms $\mp 2(\dot{x} - \Omega y)(\dot{y} + \Omega x)\sin\Omega t\cos\Omega t$ cancel and $\cos^2\Omega t + \sin^2\Omega t = 1$, so

$$\dot{X}^2 + \dot{Y}^2 = (\dot{x} - \Omega y)^2 + (\dot{y} + \Omega x)^2.$$

Expand the squares:

$$(\dot{x} - \Omega y)^2 + (\dot{y} + \Omega x)^2 = \dot{x}^2 - 2\Omega y\dot{x} + \Omega^2 y^2 + \dot{y}^2 + 2\Omega x\dot{y} + \Omega^2 x^2,$$
$$\text{so} \qquad \dot{X}^2 + \dot{Y}^2 = \dot{x}^2 + \dot{y}^2 + 2\Omega(x\dot{y} - y\dot{x}) + \Omega^2(x^2 + y^2).$$

Putting $\frac{1}{2}m$ times this into (1.6) turns

$$\begin{aligned}\mathcal{L} &= \tfrac{1}{2}m\big(\dot{X}^2 + \dot{Y}^2\big) - U(X, Y, t)\\ &= \tfrac{1}{2}m\big(\dot{X}^2 + \dot{Y}^2\big) + \frac{GmM_1}{\sqrt{(X - X_1(t))^2 + (Y - Y_1(t))^2}} + \frac{GmM_2}{\sqrt{(X - X_2(t))^2 + (Y - Y_2(t))^2}}\end{aligned}$$

into

$$\boxed{\mathcal{L} = \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + m\Omega\,(x\dot{y} - y\dot{x}) + \tfrac{1}{2}m\Omega^2(x^2 + y^2) - U(x, y)} \tag{1.10}$$

$$\mathcal{L} = \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + m\Omega\,(x\dot{y} - y\dot{x}) + \tfrac{1}{2}m\Omega^2(x^2 + y^2) + \frac{GmM_1}{\sqrt{(x + r_1)^2 + y^2}} + \frac{GmM_2}{\sqrt{(x - r_2)^2 + y^2}},$$

which is (1.1). Two remarks before differentiating: First, although it has an extra term or two in the T part, (1.10) is a big win as there is no explicit t in $\mathcal{L}$! Whew. That was the point of the coordinate change. Second, the coordinates are not "natural" in Section 7.3's sense: the relation between (x, y) and (X, Y) involves t, so T is not a pure quadratic in $\dot{x}, \dot{y}$ (i.e. there are two new terms, a linear $m\Omega(x\dot{y} - y\dot{x})$ and a velocity-free term $\frac{1}{2}m\Omega^2(x^2 + y^2)$), exactly as for the bead on the spinning hoop.

Step 5: Lagrange's equations. Not as bad as you might expect after all of that. From (1.10),

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m(\dot{x} - \Omega y), \qquad \frac{\partial \mathcal{L}}{\partial x} = m\Omega\dot{y} + m\Omega^2 x - \frac{\partial U}{\partial x},$$
$$\frac{\partial \mathcal{L}}{\partial \dot{y}} = m(\dot{y} + \Omega x), \qquad \frac{\partial \mathcal{L}}{\partial y} = -m\Omega\dot{x} + m\Omega^2 y - \frac{\partial U}{\partial y}.$$

Remembering that Ω is constant,

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{d}{dt}\, m(\dot{x} - \Omega y) = m\ddot{x} - m\Omega\dot{y}, \qquad \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{y}} = \frac{d}{dt}\, m(\dot{y} + \Omega x) = m\ddot{y} + m\Omega\dot{x},$$

so $\frac{d}{dt}\partial\mathcal{L}/\partial\dot{q}_i = \partial\mathcal{L}/\partial q_i$ for $q_i = x$ and y read

$$m\ddot{x} - m\Omega\dot{y} = m\Omega\dot{y} + m\Omega^2 x - \frac{\partial U}{\partial x},$$
$$m\ddot{y} + m\Omega\dot{x} = -m\Omega\dot{x} + m\Omega^2 y - \frac{\partial U}{\partial y}.$$

Move the $-m\Omega\dot{y}$ and the $+m\Omega\dot{x}$ from the left to the right:

$$m\ddot{x} = -\frac{\partial U}{\partial x} + m\Omega^2 x + 2m\Omega\dot{y}, \qquad m\ddot{y} = -\frac{\partial U}{\partial y} + m\Omega^2 y - 2m\Omega\dot{x}, \tag{1.11}$$

Newton's second law with three forces. The first is gravity, $-\boldsymbol{\nabla} U$, the two pulls of Step 2 written in the rotating coordinates.

The second is a centrifugal force $m\Omega^2(x, y) = m\Omega^2\, \mathbf{r}$, one force pointing straight away from the CM, where the rotation axis (the Z axis, out of the page in Figure 1) pierces the plane, with magnitude $m\Omega^2\sqrt{x^2 + y^2}$, $m\Omega^2$ times the distance from the CM. At L_1 and L_2 in Figure 1, which lie on the x axis, it points along $+x$; at L_4 it points along the line from the CM through L_4. The x equation sees its x component $m\Omega^2 x$, the y equation its y component $m\Omega^2 y$.

The third is a Coriolis force $2m\Omega(\dot{y}, -\dot{x})$, perpendicular to the velocity. It depends on velocity, so it has no associated potential energy, but it does no work, since it is always perpendicular to $\dot{\mathbf{r}}$. Chapter 9 will derive both extra forces from Newton's second law in a rotating frame;[4] here they came out of Lagrange's equations because the coordinates rotate.

Step 6: read off results. Finally! We get two results, each basically one line of reasoning from (1.10) or (1.11).

(a) *Where can m sit still?* **(I.e. where do we want to put the JWST!)** "Sit still in the rotating frame" means $\dot{x} = \dot{y} = 0$ and $\ddot{x} = \ddot{y} = 0$. Put those into (1.11): the Coriolis terms vanish with the velocities, the left sides vanish with the accelerations, and

$$0 = -\frac{\partial U}{\partial x} + m\Omega^2 x, \qquad 0 = -\frac{\partial U}{\partial y} + m\Omega^2 y, \qquad \text{i.e.} \qquad \boldsymbol{\nabla} U_{\text{eff}} = 0, \quad U_{\text{eff}} \equiv U - \tfrac{1}{2}m\Omega^2(x^2 + y^2), \tag{1.12}$$

[4] Chapter 9 writes them as $-m\ \omega \times (\omega \times \mathbf{r})$ and $-2m\ \omega \times \dot{\mathbf{r}}$ with $\omega = \Omega\hat{\mathbf{z}}$; you can check that those give the same components.

since $m\Omega^2 x = -\partial_x\big(-\frac{1}{2}m\Omega^2(x^2+y^2)\big)$ and likewise for y. Notice the order of operations here: we set the velocities to zero in the *equations of motion*, after the variation, never in $\mathcal{L}$ itself. Here, as it happens, setting $\dot{x} = \dot{y} = 0$ in (1.1) first leaves $\mathcal{L} = -U_{\text{eff}}$ and the same condition, but that is luck, and the equations of motion are gone (no $\ddot{x}$, no Coriolis term, so no way to ask whether the point is stable). In other problems, taking the "shortcut" of setting them to zero in $\mathcal{L}$ gives wrong equations outright: e.g. if you substitute the conserved angular momentum into a central-force Lagrangian before varying, the centrifugal term comes out off by a minus sign. Another way of seeing this is that we can make a statement about the motion only after Lagrange's equations are derived; only a constraint on the *configuration* (a bead on a wire, Section 7.5) goes into the coordinates before. Figure 2 shows U_{eff} for the mass ratio $\mu = 0.1$ (an unrealistically heavy M_2, chosen so the five points separate on the page) with its five stationary points.

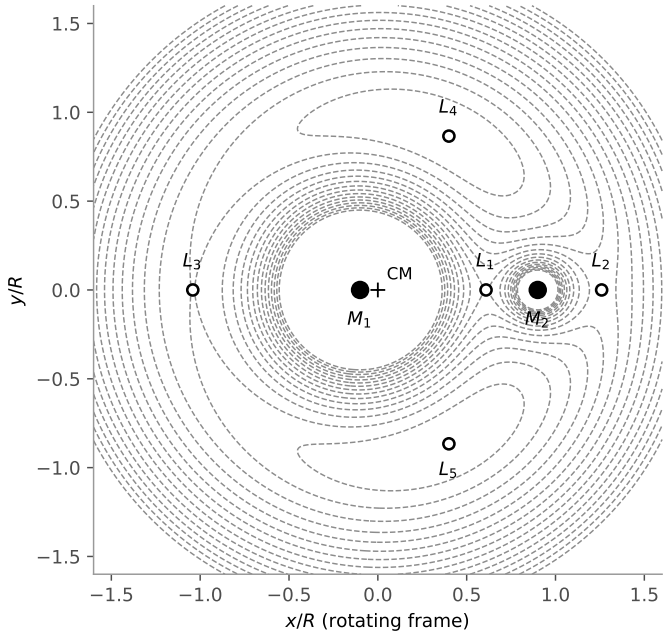


Figure 2: Contours of the effective potential U_{eff} in the rotating frame for $\mu = 0.1$, with the two primaries (filled) and the five Lagrange points (open). L_1, L_2, L_3 are saddle points; L_4 and L_5 are the two maxima. Drawn by `make_figure.py` in the handout's folder.

Now we look for solutions of $\boldsymbol{\nabla} U_{\text{eff}} = 0$. Three are on the x axis, L_1, L_2 and L_3, and they need nothing beyond (1.12): put $y = 0$, and the y equation holds by itself (every term in it carries a factor of y), while the x equation, with $d_1 = |x + r_1|$ and $d_2 = |x - r_2|$ on the axis, becomes

$$m\Omega^2 x = \frac{GmM_1\,(x + r_1)}{|x + r_1|^3} + \frac{GmM_2\,(x - r_2)}{|x - r_2|^3}, \tag{1.13}$$

the centrifugal term on the left and two copies of Newton's law of gravitation on the right, each toward its own primary. That is one equation in the one unknown x, with three roots. In Nailing down details we solve it numerically and look at the limiting case of a small μ by expanding about M_2 (see (1.17), solved in units where $R = 1$). We find that there are exactly five solutions (this is the one place we need polar coordinates; "Polar coordinates" in Nailing down details). The remaining two, L_4 and L_5, are off axis. Figure 3 shows all five.

The two off the axis are not so tricky to derive. Take the point $\mathbf{r}$ a distance R from *both* primaries, the third corner of an equilateral triangle on M_1M_2, with $\mathbf{r}_1 = (-r_1, 0)$ and $\mathbf{r}_2 = (r_2, 0)$ in the rotating frame. Then $d_1 = d_2 = R$, and the gravity term of (1.12), divided by m, is

$$\begin{aligned} -\frac{1}{m}\boldsymbol{\nabla} U &= \frac{GM_1(\mathbf{r}_1 - \mathbf{r})}{d_1^3} + \frac{GM_2(\mathbf{r}_2 - \mathbf{r})}{d_2^3} && \text{(the two pulls, Step 2)} \\ &= \frac{G}{R^3}\Big[(M_1\mathbf{r}_1 + M_2\mathbf{r}_2) - (M_1 + M_2)\,\mathbf{r}\Big] && (d_1 = d_2 = R) \\ &= -\frac{G(M_1 + M_2)}{R^3}\,\mathbf{r} && (M_1\mathbf{r}_1 + M_2\mathbf{r}_2 = 0\text{: origin at the CM}) \\ &= -\Omega^2\,\mathbf{r}\,. && \text{(by (1.4))} \end{aligned} \tag{1.14}$$

The centrifugal term of (1.12) divided by m is $+\Omega^2\,\mathbf{r}$, so the two cancel and $\boldsymbol{\nabla} U_{\text{eff}} = 0$ there. Those are L_4 and L_5, 60° ahead of and behind M_2 as seen from M_1, for any mass ratio (on M_2's orbit only in the limit $\mu \to 0$). (Jupiter's Trojans.)

In words: this is pretty neat. At the equilateral points the two bodies' gravity points exactly at the CM (look at Figure 3 and think about this: not at the center of the triangle's edge, or at the center of the triangle, but at the CM), with exactly the strength a circular orbit of angular speed Ω needs at that radius, so the test mass goes around the CM in step with the primaries!

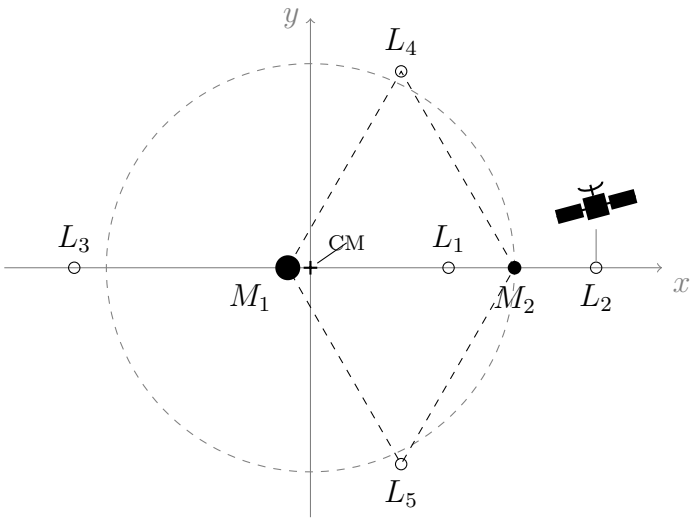


Figure 3: The five Lagrange points in the rotating frame, drawn to scale for $\mu = M_2/(M_1+M_2) = 0.1$ (the mass ratio of Figure 2; for the Sun and the Earth, L_1 and L_2 sit one percent of R from the Earth and would not separate on the page). L_1, L_2, L_3 on the line of the primaries, found from (1.17); L_4, L_5 at the third corners of the two equilateral triangles (dashed) on M_1M_2. The dashed circle is M_2's orbit about the CM. The satellite marks L_2: for the Sun and the Earth that is where JWST sits, beyond the Earth on the far side from the Sun (SOHO sits at L_1); the Trojan asteroids are at L_4 and L_5. L_1, L_2, L_3 are saddles of U_{eff} and unstable for every mass ratio, which is why JWST needs station-keeping; L_4, L_5 are maxima of U_{eff}, yet stable for $\mu < 0.0385$ ("Stability" in Nailing down details), which is why the Trojans stay put.

(b) *What is conserved?* $\mathcal{L}$ in (1.10) has no explicit t, so by Section 7.8 the Hamiltonian $\mathcal{H} =$

$\sum_i p_i\dot q_i - \mathcal{L}$ is conserved. With $p_x = \partial\mathcal{L}/\partial\dot x = m(\dot x - \Omega y)$ and $p_y = \partial\mathcal{L}/\partial\dot y = m(\dot y + \Omega x)$,

$$\begin{aligned}\mathcal{H} &= m(\dot x - \Omega y)\dot x + m(\dot y + \Omega x)\dot y - \mathcal{L} = m(\dot x^2 + \dot y^2) + m\Omega(x\dot y - y\dot x) - \mathcal{L}\\ &= m(\dot x^2 + \dot y^2) + m\Omega(x\dot y - y\dot x)\\ &\qquad - \left[\tfrac12 m(\dot x^2 + \dot y^2) + m\Omega(x\dot y - y\dot x) + \tfrac12 m\Omega^2(x^2+y^2) - U\right]\\ &= \tfrac12 m(\dot x^2 + \dot y^2) + U_{\text{eff}}(x,y),\end{aligned} \tag{1.15}$$

the last line by subtracting (1.10) term by term: the $m\Omega(x\dot y - y\dot x)$ cancel, half of $m(\dot x^2+\dot y^2)$ survives, and $-\frac12 m\Omega^2(x^2+y^2) + U$ is U_{eff}. This is the Jacobi integral (celestial mechanics usually quotes $C_J = -2\mathcal{H}/m$). Three remarks, the first two of which are the point of Section 7.8:

(a) It is conserved, and it looks like an energy: kinetic energy in the rotating frame plus the effective potential.

(b) It is *not* $T+U$. The true kinetic energy is $\frac12 m(\dot X^2 + \dot Y^2)$, and from Step 4 it is $\frac12 m(\dot x^2+\dot y^2) + m\Omega(x\dot y - y\dot x) + \frac12 m\Omega^2(x^2+y^2)$; adding U and subtracting (1.15),

$$\begin{aligned}T + U - \mathcal{H} &= \left[\tfrac12 m(\dot x^2 + \dot y^2) + m\Omega(x\dot y - y\dot x) + \tfrac12 m\Omega^2(x^2+y^2) + U\right]\\ &\qquad - \left[\tfrac12 m(\dot x^2 + \dot y^2) + U - \tfrac12 m\Omega^2(x^2+y^2)\right] = m\Omega(x\dot y - y\dot x) + m\Omega^2(x^2+y^2),\end{aligned} \tag{1.16}$$

which is not constant: in polar coordinates ($x = r\cos\theta$, $y = r\sin\theta$, so $x\dot y - y\dot x = r^2\dot\theta$ and $x^2+y^2 = r^2$) it is $m\Omega r^2\dot\theta + m\Omega^2 r^2 = \Omega\, mr^2(\dot\theta + \Omega)$, Ω times m's angular momentum about the CM in the inertial frame (whose angular velocity is $\dot\theta + \Omega$), and the primaries' pull is not aimed at the CM, so it exerts a torque. Taylor's condition for $\mathcal{H} = T + U$ (Section 7.8, pp. 269–272) was that the relation between the generalized coordinates and the inertial position not involve t; (1.7) involves t, so the condition fails, exactly as for the spinning hoop. (And $T+U$ is not conserved in the inertial frame either: there the primaries move and U depends on time. The symmetry lives in the rotating frame, and so does the conservation law.)[5]

(c) Since $\frac12 m v_{\text{rot}}^2 \geq 0$, a body with a given $\mathcal{H}$ can only be where $U_{\text{eff}} \leq \mathcal{H}$. The contours in Figure 2 are therefore fences (Hill's *zero-velocity curves*): a satellite with $\mathcal{H}$ below $U_{\text{eff}}(L_1)$ that starts near M_2 can never leave M_2's neighbourhood, however it moves. Raising $\mathcal{H}$ past $U_{\text{eff}}(L_1)$ opens a gate to M_1; past $U_{\text{eff}}(L_2)$, a gate to the outside.

That is the whole calculation. Nailing down details below adds the Hill radius (a board problem), the numbers, the polar-coordinate argument that there are exactly five points, and the question of stability, which needs the Coriolis term kept.

Nailing down details: the Hill radius, the numbers, exactly five points, stability

Board problem: L_1 and L_2 for a small μ. Use units with $R = 1$ and $G(M_1+M_2) = 1$, so $\Omega = 1$ by (1.4), $M_1 \to 1-\mu$ at $x = -\mu$ and $M_2 \to \mu$ at $x = 1-\mu$. In these units the axis equation (1.13), per unit mass, is

$$x = \frac{(1-\mu)(x+\mu)}{|x+\mu|^3} + \frac{\mu\,(x-1+\mu)}{|x-1+\mu|^3}. \tag{1.17}$$

[5] We note that this is subtle, and often interpreted incorrectly, even in published articles.

Show that for $\mu \ll 1$ both L_1 and L_2 sit a distance

$$d \approx R\left(\frac{\mu}{3}\right)^{1/3} \tag{1.18}$$

from M_2, one on each side, and evaluate it for the Sun and the Earth ($\mu = 3.00 \times 10^{-6}$, $R = 1.496 \times 10^8$ km). This d is called the *Hill radius*; inside it M_2's gravity wins over M_1's tidal pull plus the centrifugal force (the -2δ and the $+\delta$ in the solution below), which is why the Moon is Earth's.

Solution. Take L_2 first: $x = 1 - \mu + \delta$ with $\delta > 0$ (beyond M_2), so $x + \mu = 1 + \delta$ and $x - 1 + \mu = \delta$. Keep first order in δ, drop μ wherever it is added to a number of order one, but keep μ/δ^2, which is the term that matters. (This is consistent because the answer has $\delta \sim \mu^{1/3}$: then $\mu \ll \delta$, while $\mu/\delta^2 \sim \delta$.) Term by term in (1.17):

$$\begin{aligned} x = 1 - \mu + \delta &\approx 1 + \delta && \text{(the left side)} \\ \frac{(1-\mu)(x+\mu)}{|x+\mu|^3} = \frac{1-\mu}{(1+\delta)^2} &\approx 1 - 2\delta && \text{(binomial expansion } (1+\delta)^{-2} \approx 1 - 2\delta;\ 1-\mu \to 1) \\ \frac{\mu\,(x-1+\mu)}{|x-1+\mu|^3} = \frac{\mu\,\delta}{\delta^3} &= \frac{\mu}{\delta^2}. \end{aligned}$$

So

$$1 + \delta \approx 1 - 2\delta + \frac{\mu}{\delta^2} \quad \Rightarrow \quad 3\delta = \frac{\mu}{\delta^2} \quad \Rightarrow \quad \delta^3 = \frac{\mu}{3}. \tag{1.19}$$

For L_1 put $x = 1 - \mu - \delta$, so $x + \mu = 1 - \delta$ and $x - 1 + \mu = -\delta$:

$$\begin{gathered} x \approx 1 - \delta, \qquad \frac{1-\mu}{(1-\delta)^2} \approx 1 + 2\delta, \qquad \frac{\mu\,(-\delta)}{\delta^3} = -\frac{\mu}{\delta^2}, \\ 1 - \delta \approx 1 + 2\delta - \frac{\mu}{\delta^2} \quad \Rightarrow \quad -3\delta = -\frac{\mu}{\delta^2}, \end{gathered}$$

the same δ. To this order L_1 and L_2 are mirror images about M_2, and with R restored, $d = \delta R = R(\mu/3)^{1/3}$, which is (1.18). For the Sun and the Earth, $(\mu/3)^{1/3} = (1.00 \times 10^{-6})^{1/3} = 0.0100$, so $d = 0.0100$ AU $= 1.50 \times 10^6$ km, about four times the Moon's distance. JWST orbits around L_2 (a "halo orbit" a few hundred thousand km across), so that it always has the Sun, the Earth and the Moon on the same side, behind its sunshield. □

The exact positions. Equation (1.17) is a quintic once the absolute values are resolved, so the three collinear points are found numerically. There is exactly one in each of the intervals below: for f in the code (left side of (1.17) minus right side), $f'(x) = 1 + 2(1-\mu)/|x+\mu|^3 + 2\mu/|x-1+\mu|^3 > 0$, and f runs from $-\infty$ to $+\infty$ across each interval. Solving it with `scipy.optimize.brentq` on the three intervals (left of M_1, between, right of M_2):

	μ	L_1: inside M_2	L_2: beyond M_2	d from (1.18)	L_3, x/R
Sun–Earth	3.00×10^{-6}	1.49×10^6 km	1.50×10^6 km	1.50×10^6 km	-1.000001
Sun–Jupiter	9.54×10^{-4}	5.19×10^7 km	5.43×10^7 km	5.31×10^7 km	-1.0004
Earth–Moon	1.22×10^{-2}	5.80×10^4 km	6.45×10^4 km	6.13×10^4 km	-1.005

The approximation (1.18) is good to about a third of a percent for the Earth and to about five percent for the Moon, whose μ is not so small (the next term in the expansion is of relative size $\mp\frac{1}{3}(\mu/3)^{1/3}$). L_3 sits on the far side of M_1, essentially on M_2's orbit. The code is a few lines:

```
from scipy.optimize import brentq
def f(x, mu):
    return x - (1-mu)*(x+mu)/abs(x+mu)**3 - mu*(x-1+mu)/abs(x-1+mu)**3
mu = 3.003e-6
L3 = brentq(f, -2.0, -mu-1e-9, args=(mu,))
L1 = brentq(f, -mu+1e-9, 1-mu-1e-9, args=(mu,))
L2 = brentq(f, 1-mu+1e-9, 2.5, args=(mu,))
```

Polar coordinates, and why there are exactly five points. Put the test mass at (r, θ) from the CM, with θ measured from the CM-to-M_2 line, so the distances to the primaries are $r_1' = \sqrt{r^2 + 2rr_1\cos\theta + r_1^2}$ and $r_2' = \sqrt{r^2 - 2rr_2\cos\theta + r_2^2}$ (law of cosines; M_1 is at angle π, M_2 at angle 0) with $r_1 = \mu R$, $r_2 = (1-\mu)R$. The centrifugal term depends on r alone, so the θ half of (1.12) is $\partial U/\partial\theta = 0$. Differentiate $r_1'^2 = r^2 + 2rr_1\cos\theta + r_1^2$ with respect to θ, and $r_2'^2$ likewise:

$$2r_1'\frac{\partial r_1'}{\partial\theta} = -2rr_1\sin\theta, \qquad \frac{\partial r_1'}{\partial\theta} = -\frac{rr_1\sin\theta}{r_1'},$$
$$2r_2'\frac{\partial r_2'}{\partial\theta} = +2rr_2\sin\theta, \qquad \frac{\partial r_2'}{\partial\theta} = +\frac{rr_2\sin\theta}{r_2'}.$$

Then, with $U = -GmM_1/r_1' - GmM_2/r_2'$,

$$\begin{aligned}\frac{\partial U}{\partial\theta} &= \frac{GmM_1}{r_1'^2}\frac{\partial r_1'}{\partial\theta} + \frac{GmM_2}{r_2'^2}\frac{\partial r_2'}{\partial\theta} && \text{(chain rule)}\\ &= -\frac{GmM_1\, rr_1\sin\theta}{r_1'^3} + \frac{GmM_2\, rr_2\sin\theta}{r_2'^3}\\ &= Gm\,r\,M_1r_1\sin\theta\left(\frac{1}{r_2'^3} - \frac{1}{r_1'^3}\right) = 0. && (M_1r_1 = M_2r_2, \text{ the CM condition of Step 1}) \quad (1.20)\end{aligned}$$

So every equilibrium has $\sin\theta = 0$ (on the axis: L_1, L_2, L_3, one in each of the three stretches the primaries cut the axis into, as "The exact positions" above shows) or $r_1' = r_2'$ (on the perpendicular bisector of the primaries: L_4, L_5), and nothing else. On the bisector $r_1' = r_2' \equiv s$, and the algebra of Step 6(a) goes through with R replaced by s: gravity per unit mass is $-G(M_1+M_2)\,\mathbf{r}\,/s^3$, pointing at the CM, against the centrifugal $+\Omega^2\,\mathbf{r} = +G(M_1+M_2)\,\mathbf{r}\,/R^3$. So the r half of (1.12) holds only at $s = R$, the equilateral point. This factorization is the quickest way to see that there are exactly five points. (It is how Edlund, Am. J. Phys. **92**, 414 (2024), organizes the whole problem.)

Stability, in one paragraph (not derived today). L_1, L_2 and L_3 are saddle points of U_{eff} and are unstable: JWST fires its thrusters every few weeks to stay near L_2. L_4 and L_5 are *maxima* of U_{eff}, and yet they are stable whenever $27\mu(1-\mu) < 1$ (a quoted result, Gascheau's of 1843; Murray and Dermott [2], Chapter 3, derive it), i.e. $\mu < 0.0385$, which covers Sun–Jupiter (the Trojans) and Earth–Moon. A stable equilibrium at a potential maximum is impossible for a static potential: the small-oscillation analysis of Taylor's Example 7.7 would call it unstable. What rescues it is the Coriolis force, which does no work but keeps turning the velocity, so a body that starts to fall off the hill is swung around it instead (Greenberg and Davis, "Stability at potential maxima," Am. J. Phys. **46**, 1068 (1978)). Showing that means linearizing (1.11) *with* the velocity terms kept, a 4×4 eigenvalue problem; it is a very good project, and the same velocity-coupled linear equations appear in Chapter 9's Foucault pendulum.

2 The cosmological redshift

This section does the redshift problem in the order of Section 7.5's recipe (p. 258), the order Section 1 used for the Lagrange points: set the problem up, write the Lagrangian with nothing left out, state the constraints, write the Lagrangian with the constraints in, take Lagrange's equation, read off what it says.

Hubble's law and comoving coordinates. Hubble's law says a galaxy at distance r recedes from us at $v = (\dot{a}/a)\, r$, where $a(t)$ is the *scale factor*: every cosmological distance grows as $r(t) = a(t)\, x$, with x a label that does not change. Coordinates that expand with the universe like this are called *comoving coordinates*: a galaxy that just rides the expansion (the "Hubble flow") sits at a fixed x forever. A galaxy that does not just ride the flow has $r = a(t)\, x(t)$ with x changing; differentiating, $\dot{r} = \dot{a}\, x + a\, \dot{x}$, the Hubble flow $\dot{a}\, x = (\dot{a}/a)\, r$ plus the *peculiar velocity* $v_{\text{pec}} \equiv a\dot{x}$, the motion relative to the flow (a galaxy falling toward a cluster, say, or a photon crossing the universe).

The result, first. For a particle moving freely through a homogeneous expanding universe, in comoving coordinates,

$$\boxed{\mathcal{L} = \tfrac{1}{2}\, m\, a(t)^2\, \dot{x}^2} \qquad \text{(up to a total time derivative, which changes nothing; see Step 4).} \quad (2.1)$$

Two things follow immediately from Taylor's Sections 7.6 (ignorable coordinates) and 7.8 (the Hamiltonian); both are cases of Noether's theorem, that each symmetry of $\mathcal{L}$ gives a conserved quantity:

- x does not appear in $\mathcal{L}$, so Lagrange's equation for x, $\frac{d}{dt}\,\partial\mathcal{L}/\partial\dot{x} = \partial\mathcal{L}/\partial x = 0$, reads $\frac{d}{dt}(ma^2\dot{x}) = 0$: the coordinate is ignorable and $p \equiv \partial\mathcal{L}/\partial\dot{x} = ma^2\dot{x}$ is conserved. In terms of the peculiar velocity, $p = ma\,(a\dot{x}) = m\, a\, v_{\text{pec}}$ is constant, so

$$v_{\text{pec}} \propto \frac{1}{a}. \quad (2.2)$$

  Random motions die away as the universe grows ("Hubble drag"): when the universe doubles in size, every freely coasting peculiar velocity halves.

- t *does* appear in $\mathcal{L}$, through $a(t)$, so there is no conserved Hamiltonian and no conserved energy (for any particle with $v_{\text{pec}} \neq 0$; one riding the flow has $\mathcal{H} = 0$ throughout). The expanding universe does not have time-translation symmetry, and Noether's theorem has nothing to give.

Steps 1–4 are where (2.1) comes from; Step 6 is what it says about light.

Step 1: set up the problem. Fill space with matter of uniform density $\rho(t)$, expanding so that every distance grows by the same factor $a(t)$ (the scale factor). Pick any point as the origin and a test particle of mass m at position $\mathbf{r} = (X, Y, Z)$, a distance $r = |\,\mathbf{r}\,|$ from it. Newton's shell theorem (the matter outside the sphere of radius r exerts no net force; the matter inside pulls as if all of it sat at the origin; that the outside cancels is the one place Newtonian cosmology has to wave its hands, and general relativity makes the argument honest) and Newton's law of gravitation give the force on the particle, toward the origin,

$$F = -\frac{GmM(<r)}{r^2} = -\frac{Gm}{r^2}\cdot\frac{4\pi}{3}\rho r^3 = -\frac{4\pi G}{3}\, m\rho\, r, \qquad M(<r) = \tfrac{4\pi}{3}\rho r^3 \text{ the mass inside radius } r.$$

So, from $F = -\partial U/\partial r$ with $U = 0$ at the origin,

$$U(r) = -\int_0^r F\,dr' = \frac{4\pi G}{3}\,m\rho\int_0^r r'\,dr' = \frac{4\pi G}{3}\,m\rho\,\frac{r^2}{2}\,:$$

its potential energy is

$$U(r,t) = m\Phi(r,t), \qquad \Phi = \frac{2\pi G}{3}\,\rho(t)\,r^2 \qquad \Big(\text{check: } -\frac{\partial U}{\partial r} = -\frac{4\pi G}{3}\,m\rho\,r = -\frac{GmM(<r)}{r^2}\Big). \tag{2.3}$$

The expansion itself obeys the same force law. A particle riding the expansion sits at $r = a(t)\,x$ with x fixed, so $\ddot r = \ddot a\,x$, and Newton's second law for that particle reads

$$\begin{aligned} m\ddot r &= -\frac{4\pi G}{3}\,m\rho\,r \\ \ddot a\,x &= -\frac{4\pi G}{3}\,\rho\,ax && (\ddot r = \ddot a\,x \text{ and } r = ax, \text{ divided by } m) \\ \ddot a &= -\frac{4\pi G}{3}\,\rho\,a, && (\text{cancel } x) \end{aligned} \tag{2.4}$$

the Friedmann acceleration equation for a universe of pressureless matter (general relativity's version has extra terms for pressure and for the cosmological constant; with those set to zero it is exactly this). It is the one fact about the background we will use.

Step 2: the Lagrangian, with nothing left out. One particle with three Cartesian coordinates and no constraint yet, so as in Section 7.1, $\mathcal{L} = T - U$ with $r^2 = X^2 + Y^2 + Z^2$ in (2.3):

$$\mathcal{L} = \tfrac{1}{2}m\big(\dot X^2 + \dot Y^2 + \dot Z^2\big) - \frac{2\pi G}{3}\,m\rho(t)\,\big(X^2 + Y^2 + Z^2\big). \tag{2.5}$$

Lagrange's equations are Newton's second law: for X, $\partial\mathcal{L}/\partial\dot X = m\dot X$ and $\partial\mathcal{L}/\partial X = -\frac{2\pi G}{3}m\rho\cdot 2X = -\frac{4\pi G}{3}m\rho\,X$, so $\frac{d}{dt}\partial\mathcal{L}/\partial\dot X = \partial\mathcal{L}/\partial X$ reads $m\ddot X = -\frac{4\pi G}{3}m\rho\,X$, and likewise for Y and Z; together, $m\,\ddot{\mathbf{r}} = -\frac{4\pi G}{3}m\rho\,\mathbf{r}$. The time t appears explicitly through $\rho(t)$, so Section 7.8's guarantee of a constant $\mathcal{H}$ does not apply (Step 6(b) shows $\mathcal{H}$ does change), and no coordinate is yet ignorable, since U depends on all three. (The force is central, so the angular momentum about the origin is conserved; no Cartesian coordinate shows that, but Step 3 uses it.)

Step 3: the constraints. Two things are being held fixed, the same two kinds as in Section 1.

(i) *The background is prescribed.* $\rho(t)$ and $a(t)$ are given functions, not coordinates to be varied: the test particle does not change the universe it moves in (as the primaries' orbit is prescribed in the Lagrange-point problem, and the hoop's rotation in Example 7.6). This is what put t into $\mathcal{L}$.

(ii) *Motion along one line.* The particle has three coordinates; we use one. This is not a constraint in Section 7.3's sense, since nothing holds the particle on the line. It is a choice of initial conditions that the equations of motion respect: the force $-\frac{4\pi G}{3}m\rho\,\mathbf{r}$ is along $\mathbf{r}$, so a particle whose velocity is along $\mathbf{r}$ keeps its velocity along $\mathbf{r}$ (the force is central, so it exerts no torque about the origin; the particle's angular momentum $m\,\mathbf{r}\times\dot{\mathbf{r}}$ is therefore conserved, and for such a particle it starts at zero). We look only at those motions.

So once the background is prescribed, the particle has three degrees of freedom; restricting to the radial motions of (ii) (put the line along the X axis, so $Y = Z = 0$) removes two coordinates and does nothing to X. One generalized coordinate remains.

Step 4: the Lagrangian with the constraints in, in good coordinates. With $Y = Z = 0$ the Lagrangian (2.5) loses its $\dot{Y}$ and $\dot{Z}$ terms and U loses Y and Z; writing r for X (the signed distance from the origin along the line),

$$\mathcal{L} = \tfrac{1}{2} m \dot{r}^2 - \frac{2\pi G}{3} m \rho(t)\, r^2. \tag{2.6}$$

The only question left is which coordinate to use for the line. The t in (2.6) is there because the background expands, so use a coordinate that expands with it: the comoving coordinate $x = r/a(t)$, a time-dependent coordinate change of the kind Section 7.3 allows (Example 7.6 again). Then $r = ax$ and, by the product rule,

$$\dot{r} = \dot{a}\, x + a\, \dot{x}, \tag{2.7}$$

the Hubble flow $\dot{a}\, x = (\dot{a}/a)\, r$ (Hubble's law: the whole velocity of a particle riding the expansion, $\dot{x} = 0$) plus the peculiar velocity $v_{\text{pec}} \equiv a\dot{x}$, the motion relative to the flow. Put $r = ax$ and (2.7) into (2.6) and expand the square:

$$\mathcal{L} = \tfrac{1}{2} m (\dot{a}x + a\dot{x})^2 - \frac{2\pi G}{3} m \rho\, a^2 x^2 = \tfrac{1}{2} m a^2 \dot{x}^2 + m a \dot{a}\, x\dot{x} + \tfrac{1}{2} m \dot{a}^2 x^2 - \frac{2\pi G}{3} m \rho\, a^2 x^2. \tag{2.8}$$

The cross term is the one to tame. By the product rule,

$$\begin{aligned} \frac{d}{dt}\Big(\tfrac{1}{2} m a \dot{a}\, x^2\Big) &= \tfrac{1}{2} m \,\frac{d}{dt}(a\dot{a})\, x^2 + \tfrac{1}{2} m a \dot{a}\, \frac{d}{dt} x^2 \\ &= \tfrac{1}{2} m (\dot{a}^2 + a\ddot{a})\, x^2 + m a \dot{a}\, x\dot{x}, \qquad \text{(product rule again; chain rule, } \tfrac{d}{dt} x^2 = 2x\dot{x}) \end{aligned}$$

so, solving for the cross term,

$$m a \dot{a}\, x\dot{x} = \frac{d}{dt}\Big(\tfrac{1}{2} m a \dot{a}\, x^2\Big) - \tfrac{1}{2} m \dot{a}^2 x^2 - \tfrac{1}{2} m a \ddot{a}\, x^2. \tag{2.9}$$

Put (2.9) into (2.8):

$$\mathcal{L} = \tfrac{1}{2} m a^2 \dot{x}^2 + \frac{d}{dt}\Big(\tfrac{1}{2} m a \dot{a}\, x^2\Big) - \tfrac{1}{2} m \dot{a}^2 x^2 - \tfrac{1}{2} m a \ddot{a}\, x^2 + \tfrac{1}{2} m \dot{a}^2 x^2 - \frac{2\pi G}{3} m \rho\, a^2 x^2. \tag{2.10}$$

The two $\tfrac{1}{2} m \dot{a}^2 x^2$ terms cancel. By (2.4), $\ddot{a} = -\frac{4\pi G}{3} \rho\, a$, so

$$-\tfrac{1}{2} m a \ddot{a}\, x^2 = -\tfrac{1}{2} m a \Big(-\frac{4\pi G}{3} \rho\, a\Big) x^2 = +\frac{2\pi G}{3} m \rho\, a^2 x^2,$$

which cancels the potential term exactly. What is left is

$$\mathcal{L} = \tfrac{1}{2} m a^2 \dot{x}^2 + \frac{dF}{dt}, \qquad F \equiv \tfrac{1}{2} m a \dot{a}\, x^2. \tag{2.11}$$

A total time derivative added to $\mathcal{L}$ changes nothing: it adds $\int_{t_1}^{t_2} (dF/dt)\, dt = F(t_2) - F(t_1)$ to the action, and since $F = \tfrac{1}{2} m a \dot{a}\, x^2$ depends only on x and t, which Hamilton's principle (Section 7.1) holds fixed at the endpoints, that number is the same for every path with the same endpoints, so the stationary path is the same (Problem 7.48 asks you to check this at the level of Lagrange's equations. The check: put $\mathcal{L} = dF/dt = \partial_t F + \dot{x}\, \partial_x F$ into Lagrange's equation by itself: $\partial \mathcal{L}/\partial \dot{x} = \partial_x F$, so $\frac{d}{dt}\, \partial \mathcal{L}/\partial \dot{x} = \partial_t \partial_x F + \dot{x}\, \partial_x^2 F$, which is exactly $\partial \mathcal{L}/\partial x = \partial_x \partial_t F + \dot{x}\, \partial_x^2 F$ (mixed partial derivatives are equal), and the equation reads $0 = 0$.) So

$$\boxed{\mathcal{L} = \tfrac{1}{2}\, m\, a(t)^2\, \dot{x}^2} \tag{2.12}$$

which is (2.1). Two remarks before differentiating. The coordinate x is not "natural" in Section 7.3's sense: its relation to r involves t, and $\mathcal{L}$ carries t through $a(t)$. And x itself no longer appears: all of U cancelled, through (2.4), against the $a\ddot{a}$ term that the coordinate change produced. Had there been lumps in the universe, a galaxy cluster say, (2.12) would pick up $-m\phi(x,t)$ with ϕ the *peculiar* potential of the lumps alone; that is Peebles's [4] starting point for structure formation. The smooth background's potential vanished because the expansion already accounts for it.

Step 5: Lagrange's equation. From (2.12), $\partial\mathcal{L}/\partial\dot{x} = ma^2\dot{x}$ and $\partial\mathcal{L}/\partial x = 0$, so $\frac{d}{dt}\partial\mathcal{L}/\partial\dot{x} = \partial\mathcal{L}/\partial x$ reads

$$\frac{d}{dt}\big(ma^2\dot{x}\big) = 0. \tag{2.13}$$

Written out,

$$ma^2\ddot{x} + 2ma\dot{a}\,\dot{x} = 0 \quad \text{(product rule)} \qquad \Rightarrow \qquad \ddot{x} = -2\,\frac{\dot{a}}{a}\,\dot{x} \quad \text{(divided by } ma^2\text{)}:$$

in comoving coordinates, while the universe expands ($\dot{a} > 0$), a free particle decelerates at a rate set by the expansion.

Step 6: read off results. Two, each one line from (2.12) or (2.13).

(a) *What is conserved?* x is ignorable (Section 7.6), so its generalized momentum $p \equiv \partial\mathcal{L}/\partial\dot{x} = ma^2\dot{x}$ is conserved; that is what (2.13) says. In terms of the peculiar velocity, $p = ma\,(a\dot{x}) = ma\,v_{\rm pec}$, and p is constant, so

$$v_{\rm pec} \propto \frac{1}{a}: \tag{2.14}$$

when the universe doubles in size, every freely coasting peculiar velocity halves. The argument assumed $T = \frac{1}{2}mv^2$, so by itself it says nothing about light. General relativity gives the same result for any freely moving particle, massive or not: the momentum conjugate to x is conserved, and the momentum a local observer measures is that conserved momentum divided by one factor of a (here $p/a = mv_{\rm pec}$; the factor converts the comoving x to a proper distance). For a photon the measured momentum is h/λ, so $h/\lambda \propto 1/a$, $\lambda \propto a$, and

$$1 + z \equiv \frac{\lambda_{\rm obs}}{\lambda_{\rm emit}} = \frac{a_{\rm obs}}{a_{\rm emit}}, \tag{2.15}$$

the cosmological redshift. A photon emitted when the universe was half its present size arrives with twice its wavelength: $z = 1$.

(b) *What is not conserved?* $\mathcal{L}$ depends on t through $a(t)$, so Section 7.8's guarantee of a constant Hamiltonian does not apply, and the general result below shows that $\mathcal{H}$ does change. With $p = ma^2\dot{x}$,

$$\mathcal{H} = p\dot{x} - \mathcal{L} = ma^2\dot{x}^2 - \tfrac{1}{2}ma^2\dot{x}^2 = \tfrac{1}{2}ma^2\dot{x}^2 = \tfrac{1}{2}mv_{\rm pec}^2, \tag{2.16}$$

the kinetic energy in the peculiar velocity. $\mathcal{L}$ depends on t only through a, so by the chain rule

$$\frac{\partial\mathcal{L}}{\partial t} = \tfrac{1}{2}m\,(2a\dot{a})\,\dot{x}^2 = ma\dot{a}\,\dot{x}^2,$$

and Section 7.8's general result $d\mathcal{H}/dt = -\partial\mathcal{L}/\partial t$ gives

$$\begin{aligned} \frac{d\mathcal{H}}{dt} &= -ma\dot{a}\,\dot{x}^2 = -2\,\frac{\dot{a}}{a}\cdot\tfrac{1}{2}ma^2\dot{x}^2 = -2\,\frac{\dot{a}}{a}\,\mathcal{H} \\ \frac{d\ln\mathcal{H}}{dt} &= -2\,\frac{d\ln a}{dt} \quad \Rightarrow \quad \ln\mathcal{H} = -2\ln a + \text{const} \quad \Rightarrow \quad \mathcal{H} \propto \frac{1}{a^2}. \end{aligned} \tag{2.17}$$

That is consistent with (2.14): $\mathcal{H} = \frac{1}{2}mv_{\rm pec}^2$ and $v_{\rm pec} \propto 1/a$. The energy is not conserved, and the formalism says exactly why: the expanding universe has no time-translation symmetry, so Noether's theorem has nothing to give. That is the whole calculation, except for one question.

Where does the photon's energy go? Nowhere. $E = pc \propto 1/a$ simply decreases. The question presumes a conserved total energy, and Noether's theorem says where that would have to come from: time-translation symmetry. An expanding universe does not have it, so there is no global energy to balance the books with. (Locally, in any small freely-falling laboratory, energy is conserved as usual; it is the bookkeeping across cosmological distances that has no ledger.) Compare Taylor's Section 7.8, as Monday's slide put it: "if $\mathcal{L}$ does not depend explicitly on time, then energy is conserved." That statement run the other way, t in $\mathcal{L}$ so no conserved energy, is not a logical consequence of it and needs its own reason. The reason is the general result used in Step 6(b), $d\mathcal{H}/dt = -\partial\mathcal{L}/\partial t$, and here $\partial\mathcal{L}/\partial t \neq 0$. The whole universe is the example.

3 The redshift as a homework problem

The problem of Section 2, posed the way the course's optional challenge problems are posed: the setup is given, and parts (a)–(f) walk the same road, from $\mathcal{L}(r, \dot{r}, t)$ to the redshift. The solutions follow the statement (the student copy leaves them out).

Challenge problem (optional): a free particle in an expanding universe. In Newtonian cosmology, space is filled with matter of uniform density $\rho(t)$, and every distance grows by one common factor: a particle that just rides the expansion sits at $r = a(t)\,x$ with x fixed, where $a(t)$ is the *scale factor* and x is its *comoving coordinate*. By Newton's shell theorem the matter inside radius r pulls a particle there as if all of it sat at the origin (and the matter outside exerts no net force), so a particle of mass m at distance r from the origin has potential energy

$$U(r,t) = \frac{2\pi G}{3}\, m\rho(t)\, r^2 \qquad \Big[\text{check: } -\frac{\partial U}{\partial r} = -\frac{Gm}{r^2}\cdot\frac{4\pi}{3}\rho r^3 = -\frac{GmM(<r)}{r^2}\Big],$$

and the scale factor obeys the Friedmann equation $\ddot{a} = -\frac{4\pi G}{3}\rho\, a$ (Newton's second law for a particle riding the expansion, with $r = ax$; you may take it as given). Now consider a particle of mass m that moves along a line through the origin but does *not* necessarily ride the expansion.

(a) Write down the Lagrangian $\mathcal{L}(r, \dot{r}, t)$. Find the Lagrange equation for r and check that it is Newton's second law with the force above. Explain why neither of our two conservation tests (an ignorable coordinate, Section 7.6; no explicit t in $\mathcal{L}$, Section 7.8) hands you a conserved quantity here.

(b) Change to the comoving coordinate $x = r/a(t)$ (a generalized coordinate whose relation to r involves t, as in Example 7.6). Show that

$$\mathcal{L} = \tfrac{1}{2}ma^2\dot{x}^2 + \frac{d}{dt}\Big(\tfrac{1}{2}ma\dot{a}\,x^2\Big).$$

Hint: expand the square in $\frac{1}{2}m\dot{r}^2$; write the cross term $ma\dot{a}\,x\dot{x}$ as a total time derivative minus what is left over; use the Friedmann equation on what is left over.

(c) Explain why the total time derivative can be dropped without changing the equation of motion (Problem 7.48, or argue directly from Hamilton's principle). From here on take $\mathcal{L} = \frac{1}{2}ma(t)^2\dot{x}^2$.

(d) Which coordinate is now ignorable? Find the conserved generalized momentum, and show that the *peculiar velocity* $v_{\text{pec}} \equiv a\dot{x}$, the particle's velocity relative to the expansion, falls as $1/a$. (When the universe doubles in size, a galaxy's random motion halves.)

(e) Find the Hamiltonian $\mathcal{H} = p\dot{x} - \mathcal{L}$. Is it $T + U$? Is it conserved? Use $d\mathcal{H}/dt = -\partial\mathcal{L}/\partial t$ (Section 7.8) to find how $\mathcal{H}$ depends on a.

(f) General relativity gives the same conservation law for light: the momentum a local observer measures is the conserved momentum divided by a. With $p_{\text{local}} = h/\lambda$, deduce the cosmological redshift $1 + z \equiv \lambda_{\text{obs}}/\lambda_{\text{emit}} = a_{\text{obs}}/a_{\text{emit}}$.

Solutions.

(a) $\mathcal{L} = T - U$ with $T = \frac{1}{2}m\dot{r}^2$:

$$\boxed{\mathcal{L}(r, \dot{r}, t) = \tfrac{1}{2}m\dot{r}^2 - \frac{2\pi G}{3}\, m\rho(t)\, r^2}$$

Euler-Lagrange time! $\partial\mathcal{L}/\partial\dot{r} = m\dot{r}$ and $\partial\mathcal{L}/\partial r = -\frac{4\pi G}{3}m\rho\, r$, so $\frac{d}{dt}\partial\mathcal{L}/\partial\dot{r} = \partial\mathcal{L}/\partial r$ reads

$$\boxed{m\ddot{r} = -\frac{4\pi G}{3}\, m\rho\, r}$$

which is $m\ddot{r} = -GmM(<r)/r^2$ with $M(<r) = \frac{4\pi}{3}\rho r^3$: Newton's second law with the shell-theorem force. Neither test applies: r appears in $\mathcal{L}$ (through U), so it is not ignorable, and t appears explicitly (through $\rho(t)$), so Section 7.8's guarantee of a constant $\mathcal{H}$ is off. (A conserved quantity does exist, as (d) shows; these coordinates just do not reveal it.)

(b) With $r = ax$, the product rule gives $\dot{r} = \dot{a}x + a\dot{x}$. Substituting into (a),

$$\mathcal{L} = \tfrac{1}{2}m(\dot{a}x + a\dot{x})^2 - \frac{2\pi G}{3}\, m\rho\, a^2x^2 = \tfrac{1}{2}ma^2\dot{x}^2 + ma\dot{a}\, x\dot{x} + \tfrac{1}{2}m\dot{a}^2x^2 - \frac{2\pi G}{3}\, m\rho\, a^2x^2.$$

The cross term: by the product rule (and the chain rule for $\frac{d}{dt}x^2 = 2x\dot{x}$),

$$\frac{d}{dt}\Big(\tfrac{1}{2}ma\dot{a}\, x^2\Big) = \tfrac{1}{2}m(\dot{a}^2 + a\ddot{a})\, x^2 + ma\dot{a}\, x\dot{x} \quad\Longrightarrow\quad ma\dot{a}\, x\dot{x} = \frac{d}{dt}\Big(\tfrac{1}{2}ma\dot{a}\, x^2\Big) - \tfrac{1}{2}m\dot{a}^2x^2 - \tfrac{1}{2}ma\ddot{a}\, x^2.$$

Put that in:

$$\mathcal{L} = \tfrac{1}{2}ma^2\dot{x}^2 + \frac{d}{dt}\Big(\tfrac{1}{2}ma\dot{a}\, x^2\Big) - \tfrac{1}{2}m\dot{a}^2x^2 - \tfrac{1}{2}ma\ddot{a}\, x^2 + \tfrac{1}{2}m\dot{a}^2x^2 - \frac{2\pi G}{3}\, m\rho\, a^2x^2.$$

The two $\frac{1}{2}m\dot{a}^2x^2$ terms cancel. Friedmann, $\ddot{a} = -\frac{4\pi G}{3}\rho\, a$, turns the $a\ddot{a}$ term into $-\frac{1}{2}ma\big(-\frac{4\pi G}{3}\rho\, a\big)x^2 = +\frac{2\pi G}{3}\, m\rho\, a^2x^2$, which cancels the potential term exactly. What is left is

$$\boxed{\mathcal{L} = \tfrac{1}{2}ma^2\dot{x}^2 + \frac{d}{dt}\Big(\tfrac{1}{2}ma\dot{a}\, x^2\Big)}$$

(The whole of U went into the $a\ddot{a}$ term: the expansion is *caused* by that gravity, so in coordinates that expand with it, the gravity is already accounted for.)

(c) Write $F = \frac{1}{2}ma\dot{a}\, x^2$, a function of x and t only. Adding dF/dt to $\mathcal{L}$ adds $\int_{t_1}^{t_2}(dF/dt)\, dt = F(t_2) - F(t_1)$ to the action, and Hamilton's principle holds $x(t_1)$ and $x(t_2)$ fixed, so that number is the same for every path being compared. The stationary path, and so the equation of motion, is unchanged. (Problem 7.48 checks the same thing at the level of Lagrange's equations: $\partial(dF/dt)/\partial x$ and $\frac{d}{dt}\partial(dF/dt)/\partial\dot{x}$ are both $\partial_x\partial_t F + \dot{x}\,\partial_x^2 F$, so they cancel.)

(d) With $\mathcal{L} = \frac{1}{2}ma^2\dot{x}^2$, x does not appear: x is ignorable, and

$$\boxed{p \equiv \frac{\partial\mathcal{L}}{\partial\dot{x}} = ma^2\dot{x} \text{ is conserved}}$$

since Lagrange's equation reads $\frac{d}{dt}(ma^2\dot{x}) = \partial\mathcal{L}/\partial x = 0$. In terms of the peculiar velocity, $p = ma\,(a\dot{x}) = ma\, v_{\text{pec}}$, and p is constant, so

$$\boxed{v_{\text{pec}} \propto \frac{1}{a}}$$

Double the universe, halve every peculiar velocity.

(e) With $p = ma^2\dot{x}$,

$$\mathcal{H} = p\dot{x} - \mathcal{L} = ma^2\dot{x}^2 - \tfrac{1}{2}ma^2\dot{x}^2 = \tfrac{1}{2}ma^2\dot{x}^2 = \boxed{\tfrac{1}{2}mv_{\text{pec}}^2}$$

the kinetic energy in the peculiar velocity. It is not $T+U$: $T = \frac{1}{2}m\dot{r}^2 = \frac{1}{2}m(\dot{a}x + a\dot{x})^2$ contains the Hubble-flow part, and U is not in $\mathcal{H}$ at all. That is allowed, because the relation $r = a(t)x$ between the generalized coordinate and the position involves t (Section 7.8's condition for $\mathcal{H} = T + U$ fails). It is not conserved either: $\mathcal{L}$ depends on t through a, so by the chain rule $\partial\mathcal{L}/\partial t = \frac{1}{2}m(2a\dot{a})\dot{x}^2 = ma\dot{a}\,\dot{x}^2$, and

$$\frac{d\mathcal{H}}{dt} = -\frac{\partial\mathcal{L}}{\partial t} = -ma\dot{a}\,\dot{x}^2 = -2\,\frac{\dot{a}}{a}\cdot\tfrac{1}{2}ma^2\dot{x}^2 = -2\,\frac{\dot{a}}{a}\,\mathcal{H} \quad\Longrightarrow\quad \boxed{\mathcal{H} \propto \frac{1}{a^2}}$$

($d\ln\mathcal{H}/dt = -2\,d\ln a/dt$ integrates to $\ln\mathcal{H} = -2\ln a + \text{const.}$) That agrees with (d): $\mathcal{H} = \frac{1}{2}mv_{\text{pec}}^2$ and $v_{\text{pec}} \propto 1/a$.

(f) The local momentum is p/a with p constant, so $p_{\text{local}} \propto 1/a$. For a photon $p_{\text{local}} = h/\lambda$, so $h/\lambda \propto 1/a$, i.e. $\lambda \propto a$, and

$$\boxed{1 + z = \frac{\lambda_{\text{obs}}}{\lambda_{\text{emit}}} = \frac{a_{\text{obs}}}{a_{\text{emit}}}}$$

A photon emitted when the universe was half its present size arrives with twice its wavelength: $z = 1$.

Assess: The two results are the same statement about the same symmetry. In comoving coordinates nothing depends on x, so the momentum conjugate to x is conserved; measured locally that momentum falls as $1/a$, which is Hubble drag for a galaxy and the redshift for a photon. Energy is not conserved, and the formalism says exactly why: t is in $\mathcal{L}$, the expanding universe has no time-translation symmetry, and $d\mathcal{H}/dt = -\partial\mathcal{L}/\partial t$ gives the rate. The one hand-wave is the shell theorem on an infinite medium; general relativity makes that honest and gives (f) without borrowing.

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